

① Find the rank of the Matrix:-

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \Rightarrow \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Rank of matrix A, $\rho(A) = \underline{\underline{2}}$

② Find the rank of matrix $A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_1 \longleftrightarrow R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & -5 & -9 & 1 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_4 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & -2 & 2 \\ 0 & -5 & -9 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -5 & -9 & 1 \end{bmatrix} =$$

$$R_3 \leftrightarrow R_4$$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -5 & -9 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 5R_2$$

$$= \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 4 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 \cdot \frac{1}{4}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore P(A) = 3$$

$$3. \quad A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 2 & 1 & 3 & 4 \\ 4 & 0 & 2 & 1 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 - R_1 \end{aligned} \Rightarrow \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & -2 & 0 \end{bmatrix}$$

$$\begin{aligned} R_3 &\rightarrow R_3 + R_2 \\ R_4 &\rightarrow R_4 + R_2 \end{aligned} = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & -6 & -7 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 2R_3 = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow \frac{R_1}{2}, \quad R_2 \rightarrow \frac{R_2}{-2}, \quad R_3 \rightarrow \frac{R_3}{-3}$$

$$A = \begin{bmatrix} 1 & 1 & 3 & 4 \\ 0 & 1 & 2 & 7/2 \\ 0 & 0 & 1 & 4/3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since there are 4 non-zero rows,
Rank of matrix, $\rho(A) = \underline{\underline{4}}$

$$4. \quad A = \begin{bmatrix} 1 & 0 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array} \Rightarrow \begin{bmatrix} 1 & 0 & -3 & -1 \\ 0 & 0 & 4 & 2 \\ 0 & 1 & 9 & 5 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\cancel{R_3} \Rightarrow R_4 \rightarrow R_4 - R_3$$

$$A = \begin{bmatrix} 1 & 0 & -3 & -1 \\ 0 & 0 & 4 & 2 \\ 0 & 1 & 9 & 5 \\ 0 & 0 & -8 & -4 \end{bmatrix}$$

Since there are 4 non-zero rows in this echelon form.

$$\therefore \rho(A) = 4$$

4. Solve :- $5x_1 + x_2 + 3x_3 = 20$

$$2x_1 + 5x_2 + 2x_3 = 18$$

$$3x_1 + 2x_2 + x_3 = 14$$

Soln: The augmented matrix $[A:B] = \left[\begin{array}{ccc|c} 5 & 1 & 3 & 20 \\ 2 & 5 & 2 & 18 \\ 3 & 2 & 1 & 14 \end{array} \right]$

$$R_2 \leftrightarrow R_1 = \begin{bmatrix} 2 & 5 & 2 & : & 18 \\ & & & & \\ & & & & \end{bmatrix} \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 2 & 5 & 2 & : & 18 \\ 3 & 2 & 1 & : & 14 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_3 \quad 5R_2 - 2R_1$$

$$R_3 \rightarrow 5R_3 - 3R_1 = \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 0 & 23 & 4 & : & 50 \\ 0 & 7 & -4 & : & 10 \end{bmatrix}$$

$$R_3 \rightarrow 23R_3 - 7R_2$$

$$= \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 0 & 23 & 4 & : & 50 \\ 0 & 0 & -120 & : & -120 \end{bmatrix}$$

$$\text{Rank of } [A] = \text{Rank of } [A:B] = 3$$

$\therefore$ System is Consistent

$$\text{Rank} = \underline{\text{no. of unknowns}}$$

$$3 = 3$$

$\therefore$ This system has unique solutions.

$$5x_1 + x_2 + 3x_3 = 20 \rightarrow (1)$$

$$23x_2 + 4x_3 = 50 \rightarrow (2)$$

$$-120x_3 = -120 \rightarrow (3)$$

$$120x_3 = 120$$

$$\boxed{x_3 = 1} \text{ substitute in (2)}$$

$$23x_2 + 4(1) = 50$$

$$23x_2 = 46$$

$$x_2 = \frac{46}{23}$$

$$\boxed{x_2 = 2}$$

$$\text{eq (1)}$$

$$5x_1 + 2 + 3 = 20$$

$$5x_1 = 15$$

$$\boxed{x_1 = 3}$$

① Test for consistency and Solve.

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$3x + y + z = 8$$

Soln:
=

The augmented matrix $[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & -1 & 2 & : & 5 \\ 3 & 1 & 1 & : & 8 \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 1 & : & -1 \\ 0 & -2 & -2 & : & -10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \Rightarrow$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 1 & : & -1 \\ 0 & 0 & -3 & : & -9 \end{bmatrix}$$

$$\text{Rank of } [A:B] = \text{Rank of } A$$

$\therefore$ The system is consistent.

Rank = no. of unknown

$$3 = 3$$

Since rank, $r = n$, we have unique solutions.

$$x + y + z = 6 \rightarrow (1)$$

$$-2y + z = -1 \rightarrow (2)$$

$$3z = -9 \rightarrow (3)$$

$(z = -3)$ substitute (3) in (2)

$$-2y + 3 = -1$$

$$\Rightarrow 2y = -4 \quad [y = -2]$$

$$x + 2 + 3 = 6$$

$$[x = 1]$$

These are the solutions

Q. Solve $x + 2y + 3z = 14$

$$4x + 5y + 7z = 35$$

$$3x + 3y + 4z = 21$$

Soln

The augmented matrix $[A:B] = \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 4 & 5 & 7 & : & 35 \\ 3 & 3 & 4 & : & 21 \end{bmatrix}$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_3 \rightarrow R_3 - 3R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & -3 & -5 & : & -21 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

The Rank of $[A:B] = \text{Rank } [A]$
 $\therefore$ The system is consistent

Rank = no. of unknowns

$$2 = 3$$

$$2 \neq 3$$

Since $r \neq n$, we have infinite ^{many} solutions.

$n - r = 3 - 2 = 1 \Rightarrow$ one arbitrary constant exist (k)

let $z =$

$$x + 2y + 3z = 14 \rightarrow (1)$$

$$-3y - 5z = -21 \rightarrow (2)$$

let $z = k$

$$-3y - 5k = -21$$

$$-3y = -21 + 5k$$

$$y = \frac{21 - 5k}{3}$$

$$\boxed{y = \frac{-5k + 21}{3}} \text{ substitute in (1)}$$

$$x + 2y + 3z = 14$$

$$x = 14 - 2y - 3z$$

$$14 - \left(\frac{-5k + 21}{3}\right) \cdot 2 - 3k$$

$$x = 14 + \frac{10k}{3} - 42 - 3k$$

$$x = \frac{42 + 10k - 42 - 9k}{3}$$

$$\boxed{x = \frac{k}{3}}$$

Thus $x = \frac{k}{3}$, $y = \frac{7 - 5k}{3}$ and $z = k$ represents infinite solⁿ since k is arbitrary.

Q) Investigate the values of λ and μ such that the system of equations

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

may have,

(i) Unique solⁿ

(ii) Infinite solⁿ

(iii) No solution:

(i) The augmented matrix $[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$= \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda - 1 & : & \mu - 6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$$

(i) Unique solⁿ when $\lambda \neq 3$ $\mu \neq 10$

then rank of matrix $[A] = \text{Rank of } [A:B] = \text{no. of unknowns} = 3$

System is consistent & since it is equal to no. of unknowns
 $\therefore$ Then system of equations will have unique solution

(ii) Infinite solution.

when $\lambda = 3$ $\mu = 10$

Rank of $[A] = \text{Rank of } [A:B] = 2$

But Rank < no. of unknowns.

System is consistent.

$\therefore$ The System of eqⁿ will have infinite solutions.

iii) No solution.

If $\lambda = 3$ and $\mu \neq 10$

Then,

Rank of $[A]$ is 2 and Rank of $[A:B] = 3$

$\therefore \rho[A] \neq \rho[A:B]$

The system inconsistent.

$\therefore$ The system of eqⁿ will have no solutions.